

Bridging Light and Radio: Hybrid Network Architectures for IoT in the 6G Era

Mohammad Khalili¹, Amila Perera¹, Juha Häkkinen, Marcos Katz¹, Konstantin Mikhaylov¹

1. Centre for Wireless Communications, University of Oulu

Abstract

The massive scale and dense deployments of future Internet of Things (IoT), particularly within the 6G framework, expose the critical limitations of traditional Radio Frequency (RF) systems, namely spectral congestion, interference, and security vulnerabilities. Optical Wireless Communication (OWC) offers a compelling complement, providing vast, unlicensed spectrum, immunity to RF interference, and enhanced physical-layer security. However, OWC's susceptibility to blockages and limited mobility hinders its standalone deployment, as it cannot, on its own, meet the stringent connectivity and mobility requirements of many IoT applications. This article explores the convergence of optical and radio networks and proposes a flexible, adaptable, and resilient connectivity framework tailored for modern IoT ecosystems. We present a holistic system architecture that integrates single-mode RF, single-mode OWC, and unified hybrid Access Points (APs) to efficiently manage the diverse and often conflicting objectives of various IoT applications. While these hybrid RF/OWC APs offer significant advantages in cost, efficiency, and coordinated management, they also introduce a primary deployment challenge: the inherent coverage mismatch between the two technologies. To precisely address this critical issue, we demonstrate a practical solution supported by a coverage equivalence analysis to validate the proposed framework. This analysis quantifies the required OWC transmitter density to match a given RF range, offering tangible guidance for system planning.

Keywords

communication, networking and broadcast technologies, hybrid network architecture, internet of things, optical wireless communication

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Mohammad Khalili*, Amila Perera*, Juha Häkkinen†, Marcos Katz*, and Konstantin Mikhaylov*

*Centre for Wireless Communications, University of Oulu, Oulu, Finland

{mohammad.khalili, malalgodage.perera, marcos.katz, konstantin.mikhaylov}@oulu.fi

† Circuits and Systems Research Unit (CAS), University of Oulu, Oulu, Finland

juha.hakkinen@oulu.fi

Abstract—The massive scale and dense deployments of future Internet of Things (IoT), particularly within the 6G framework, expose the critical limitations of traditional Radio Frequency (RF) systems, namely spectral congestion, interference, and security vulnerabilities. Optical Wireless Communication (OWC) offers a compelling complement, providing vast, unlicensed spectrum, immunity to RF interference, and enhanced physical-layer security. However, OWC’s susceptibility to blockages and limited mobility hinders its standalone deployment, as it cannot, on its own, meet the stringent connectivity and mobility requirements of many IoT applications. This article explores the convergence of optical and radio networks and proposes a flexible, adaptable, and resilient connectivity framework tailored for modern IoT ecosystems. We present a holistic system architecture that integrates single-mode RF, single-mode OWC, and unified hybrid Access Points (APs) to efficiently manage the diverse and often conflicting objectives of various IoT applications. While these hybrid RF/OWC APs offer significant advantages in cost, efficiency, and coordinated management, they also introduce a primary deployment challenge: the inherent coverage mismatch between the two technologies. To precisely address this critical issue, we demonstrate a practical solution supported by a coverage equivalence analysis to validate the proposed framework. This analysis quantifies the required OWC transmitter density to match a given RF range, offering tangible guidance for system planning.

I. INTRODUCTION

Radio Frequency (RF) communication is the foundational and most widely used wireless technology globally, forming the backbone of Internet of Things (IoT) connectivity infrastructure. While RF systems offer reliable coverage and mobility suitable for many applications, the increasing density and scale of IoT deployments, particularly within the 6G framework, have revealed critical limitations. Spectral congestion and frequency interference significantly constrain capacity and scalability in dense environments. Moreover, RF systems are susceptible to security vulnerabilities and Electromagnetic Interference (EMI). While ongoing research continues to enhance RF capabilities, alternative communication media are rapidly emerging as promising complements to address these evolving demands [1]–[3].

Optical Wireless Communication (OWC) is a strong candidate to address RF’s limitations. In particular, the Line-of-Sight (LoS) nature of optical signals restricts transmission to the immediate area of emission. This spatial confinement enables enhanced physical-layer security and efficient fre-

quency reuse across neighboring zones or floors. In addition, the optical spectrum remains largely unregulated—particularly indoors—allowing flexible network expansion, and lower interference in dense IoT scenarios. Moreover, optical signals are inherently immune to EMI, making them a safe and reliable choice for deployment in sensitive environments such as hospitals, aircraft cabins, and industrial facilities [3]. OWC encompasses technologies such as Visible Light Communication (VLC), Infrared (IR), Optical Camera Communication (OCC), Light Fidelity (LiFi), and Free Space Optics (FSO). However, OWC systems are highly vulnerable to physical obstructions, severely restricted by spatial boundaries, and exhibit limited support for device mobility compared to RF-based solutions [4].

Clearly, both RF and OWC have their respective strengths and limitations, and no single communication technology excels across all performance metrics. However, RF and OWC can complement each other, with the strengths of one effectively offsetting the fundamental limitations of the other. As a result, integrating RF and OWC within a unified network presents a practical and effective solution for next-generation wireless communication, paving the way for a more robust and complete connectivity solution [2]–[4]. Driven by these advantages, several hybrid communication strategies have been proposed in recent years, such as LiFi/Wi-Fi, VLC/Zigbee, and VLC/Wi-Fi combinations [5]. However, most hybrid solutions are designed for general-purpose wireless networks, often overlooking key IoT characteristics [5], [6].

This article demonstrates why hybrid RF/OWC networking is well-suited for the IoT ecosystem and shows how dynamic reconfigurability between the two technologies can address critical needs in energy efficiency, security, scalability, and resilience. To translate this potential into practice, this article presents four major contributions focused on the practical deployment of hybrid RF/OWC networks in IoT ecosystems. First, we introduce a unified and hierarchical system architecture engineered to balance diverse and often competing performance demands in hybrid RF/OWC networks for IoT applications. Second, we analyze five connectivity modalities for IoT ecosystems using hybrid RF/OWC networks, comparing their relative performance across ten key technical and deployment metrics. Third, we propose the integrated hybrid RF/OWC Access Point (AP) as a deployment option for establishing a hybrid zone, demonstrating its clear advantages

in cost, coordination, and manageability. Fourth, to address the central challenge of coverage mismatch between RF and OWC, we develop a coverage-equivalence analysis that provides actionable, quantitative guidelines for hybrid RF/OWC coverage. The article concludes by identifying key challenges and outlining crucial future research directions necessary for the widespread adoption of hybrid RF/OWC networks for modern IoT applications.

II. WHY HYBRID RF + LIGHT WILL POWER 6G IOT

The modern IoT environment is characterized by dense deployments of devices operating under strict energy, cost, and processing constraints. Effective solutions for IoT ecosystems must balance multiple, often competing, design objectives—such as optimal resource utilization, enhanced security, high scalability, robust resilience, and energy efficiency—all while meeting the sustainability targets for 6G and beyond. This challenge is further compounded by the wide disparity in Quality-of-Service (QoS) demands from different IoT applications, ranging from ultra-low-latency, high-reliability links in industrial automation to low-cost, energy-constrained connectivity in environmental sensing [7].

Balancing these often-conflicting objectives poses a significant challenge for traditional RF-only systems [8]. Given that OWC can effectively address part of these demands (e.g., spectral density, security) while RF satisfies others (e.g., mobility, coverage), the integration of RF and OWC in hybrid RF/OWC networks introduces the necessary flexibility and diversity to comprehensively address the heterogeneous performance requirements of modern IoT ecosystems. This section examines the core communication requirements of IoT applications and details how hybrid RF/OWC networks can address them.

1) *Energy Efficiency*: Energy is a vital resource in IoT systems and must be managed judiciously to extend device lifespans, enhance network sustainability, and reduce costs. Among OWC technologies, VLC and IR communication stand out for their potential to enhance energy efficiency in IoT ecosystems. VLC enables simultaneous illumination and data transmission to IoT devices, effectively eliminating the need for additional energy dedicated to communication. In addition, IR provides an energy-efficient option for transmissions, particularly in scenarios where illumination is unnecessary and data payloads are minimal. In addition, emerging techniques such as Simultaneous Wireless Information and Power Transfer (SWIPT) in RF systems and Simultaneous Lightwave Information and Power Transfer (SLIPT) in VLC enable devices to harvest energy while receiving data, where SLIPT has demonstrated transmission power in the range of 0.1W to 10W [3], [9].

2) *Security and Privacy*: The IoT ecosystem faces critical security vulnerabilities, yet the inherent resource constraints of most IoT devices preclude the utilization of advanced security approaches, such as computationally intensive cryptographic algorithms, defining a critical security dilemma in IoT design. Due to the spatial confinement of optical signals and their inability to penetrate opaque barriers, OWC substantially

mitigates the risk of eavesdropping and is inherently resistant to RF-specific vulnerabilities such as electromagnetic leakage and signal sniffing. Furthermore, OWC is more resistant to jamming because interference requires direct LoS and precise alignment, unlike RF, which is vulnerable to omnidirectional attacks. The spatial confinement of OWC also enables location-based access control, which facilitates an essential form of micro-segmentation and enhances device authentication by limiting the attack surface to a defined physical area. Moreover, multi-connectivity adds another layer of resilience and security. A device can split sensitive data across both RF and OWC links and encode each part separately before transmitting over the distinct channels. This data is then reassembled only at the intended receiver, reducing the risk of interception on any single channel, thereby contributing to trustworthiness across the network domains.

3) *Scalability and Coverage*: Scalability and coverage are critical to support a large number of IoT devices and ensure widespread, reliable connectivity across diverse deployment environments. OWC enhances scalability in dense wireless environments by utilizing small, confined cells and the extensive unlicensed optical spectrum, allowing multiple interference-free links to coexist in close proximity and support high IoT device density. RF complements OWC by extending network coverage, enabling Non-Line-of-Sight (NLoS) communication, and providing robust mobility support. It also benefits from mature infrastructure, globally accepted standards, and broad interoperability across devices and platforms. By combining OWC for localized, high-capacity communication and RF for broad coverage and mobility, hybrid RF/OWC systems achieve enhanced scalability and robust coverage.

4) *Reliable and Resilient Communication*: Maintaining robust communication is essential for IoT systems, particularly for mission-critical or real-time tasks under dynamic conditions. This necessity moves beyond reliability (consistent performance under typical conditions) and robustness (withstanding predicted failures) to prioritize true resilience (the ability to maintain critical functionality and rapidly recover from unexpected failures). OWC strengthens this capability by delivering low-latency, interference-free links within confined spaces. In contrast, RF technologies ensure ubiquitous coverage and mobility support necessary for continuous service. Moreover, hybrid RF/OWC systems support multi-connectivity, which is a crucial form of redundancy and diversity. This enables dynamic failover mechanisms, allowing devices to seamlessly switch between or simultaneously utilize both links based on real-time network conditions and application requirements. For example, if an OWC link is suddenly blocked, a device can maintain its critical session seamlessly over the RF link, or vice-versa if the RF channel experiences spectral congestion or EMI. This dynamic adaptability enables the network to recover from both known and unknown disruptions, thereby transforming link-level redundancy into system-level resilience [10].

5) *Localization and Positioning*: Accurate localization plays a critical role in enabling context-aware IoT applications such as asset tracking, environmental monitoring, and location-based automation. RF-based approaches often fail indoors due to severe signal attenuation and multipath propagation,

while also imposing high energy costs. Because optical signals are confined to a specific area, the reception of an OWC signal inherently confirms the presence of a device within that space, thereby providing implicit and bounded location information [11]. In addition, Visible Light Positioning (VLP) enables high-precision indoor localization; for instance, a three-dimensional mobile online VLP prototype achieved centimeter-level accuracy with an average positioning error of 1.77 cm [12].

6) *Seamless Integration into the Environment*: VLC's ability to leverage existing lighting infrastructure enables cost-effective deployment across a wide range of IoT applications, often requiring minimal modifications to the physical environment. This integration not only reduces installation complexity and lowers operational costs but also promotes sustainability by reusing existing resources. Furthermore, the inherent immunity of OWC systems to EMI makes them a safe and reliable choice in electromagnetically sensitive environments (e.g., hospitals, industrial facilities) or in scenarios where RF noise limits performance. This benefit also extends from the spatial domain (using OWC in specific locations) to the temporal domain; OWC can be strategically employed when the RF spectrum becomes congested, thus allowing the system to offload traffic and reduce competition for the crowded radio channel.

III. A TRIZONE-BASED PARTITIONING FRAMEWORK FOR HYBRID IoT NETWORKS

The modern IoT application—characterized by strict constraints, competing design objectives, and widely disparate QoS demands—necessitates a flexible and adaptive deployment strategy. A practical approach to address this diversity is to partition the deployment environment into three distinct functional zones based on communication requirements and environmental characteristics. The first is an OWC-dominated zone, ideal for confined, static areas demanding high capacity, high density environment, low interference, and strong physical-layer security. The second is an RF-dominated zone, where NLoS connectivity, mobility support, or wide-area coverage is paramount, particularly in areas with scarce device density. Finally, a hybrid zone requires RF and OWC to coexist, providing complementary coverage, enhancing reliability, and ensuring seamless communication across diverse conditions. The size and even the existence of these distinct zones, however, ultimately depend on the specific IoT application and its requirements. In many scenarios, a larger hybrid zone—or one that encompasses the entire environment—may be the optimal solution, as it best utilizes the complementary benefits of both RF and OWC technologies.

A. System Architecture

Figure 1 details a holistic system architecture, highlighting the key components of a hybrid RF/OWC network designed for diverse IoT applications. The network follows a hierarchical structure, extending from cloud services down to IoT devices. Within this hierarchy, the hybrid RF/OWC network functions as a Local Area Network (LAN) that connects to the

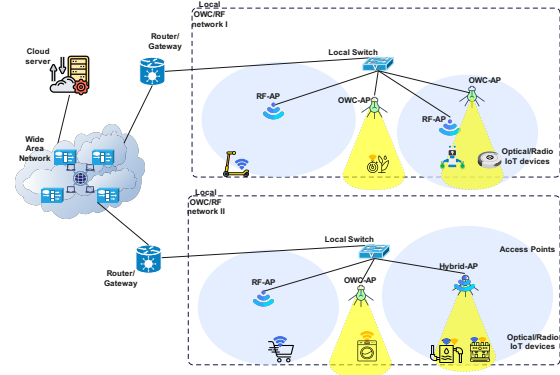


Figure 1. System architecture of a hybrid RF/OWC network for IoT applications. The zones are distinguished by their coverage availability: RF-only, OWC-only, and hybrid (both RF and OWC). The figure also illustrates two deployment models for the hybrid zone: one using separate single-mode APs and another using an integrated hybrid RF/OWC AP.

Internet or a Wide Area Network (WAN) through dedicated gateways. Multiple hybrid LANs can operate concurrently and interface with a shared cloud platform, facilitating large-scale coordination, data analytics, and service orchestration.

Similar to conventional Wireless LANs (WLANs), a local hybrid RF/OWC network comprises multiple APs that form the communication backbone between IoT devices and the broader infrastructure. These APs are typically interconnected through wired or wireless backhaul links, facilitating efficient data exchange and centralized network control.

B. Communication Modalities

In hybrid RF/OWC networks, a diverse spectrum of short-range communication technologies can be employed to establish links between IoT devices and APs. On the RF side, established standards such as Bluetooth Low Energy (BLE), Wi-Fi, and Zigbee—alongside emerging protocols like DECT NR+ and Wi-Fi HaLow—offer distinct trade-offs in range, energy consumption, and throughput [13]. Conversely, the OWC domain presents a different set of options, including VLC, IR, OCC, and LiFi. VLC exploits existing lighting infrastructure to deliver simultaneous illumination and data transmission, providing an inherently energy-efficient solution. IR, invisible to the human eye, supports secure, low-energy, point-to-point links. While LiFi could achieve 10+ Gbps data rates [14], its associated high throughput, complexity, and cost make it a mismatch for general IoT, positioning it for high-performance networking instead [4], [14]. OCC leverages image sensors for communication, but its dependence on cameras and flashes—hardware rarely found in resource-constrained IoT nodes—restricts its viability for many IoT applications.

Among the OWC candidates for IoT ecosystems, VLC is the most promising. The illumination–communication duality provides a significant advantage for the downlink, yet realizing a reliable VLC uplink channel (device-to-AP) is fundamentally problematic. The requirement for a bright, high-power light source on the device to reach the ceiling-mounted AP directly conflicts with the stringent IoT energy constraints. Beyond the

issues of increased hardware cost and energy consumption, the resultant visual disturbance from modulated light emissions, alongside increased hardware cost and energy consumption, fundamentally renders the VLC uplink impractical for human-centric or low-power IoT deployments.

To overcome VLC's limitations on the uplink, communication from devices to the AP can instead be realized using IR or RF channels, depending on system design and application requirements. IR-based uplinks, in particular, retain OWC's intrinsic benefits while eliminating visible light emission and enabling ultra-low power operation ideal for IoT devices. In certain specialized configurations, designers can implement RF downlinks paired with IR uplinks to exploit the complementary strengths of both technologies, thereby enhancing physical-layer confidentiality and ensuring robust, secure data transmission from IoT nodes.

The apex of hybrid design is the multi-connectivity modality, which supports concurrent VLC/RF downlink and IR/RF uplink. By employing both RF and OWC interfaces, the multi-connectivity capable IoT devices can harness the distinct and complementary benefits of both wireless media in parallel. This capability enables advanced and dynamic modality selection, allowing the system to meet application-specific QoS requirements under varying operating conditions [15].

Figure 2 presents a comparative assessment of the five communication modalities, highlighting their relative performance across ten key technical and deployment metrics. The multi-connected mode clearly stands out as the most capable option, offering consistently high performance across various criteria. In particular, this mode provides significant flexibility, allowing the system to dynamically adapt to the specific IoT application requirements. This flexibility means that, at a minimum, the multi-connected mode is capable of matching the performance of the most suitable single-mode link for any given metric—for instance, it can provide mobility and coverage equivalent to that of single-mode RF. Furthermore, it can surpass the performance of single-mode approaches by, for example, using both links simultaneously to provide enhanced security or support higher device density. This advanced flexibility, however, comes with trade-offs—primarily increased implementation complexity, resulting in higher hardware cost. Overall, for mission-critical IoT scenarios where reliability, throughput, and security are paramount, the superior performance and flexibility delivered by the multi-connected mode justify the added system complexity and investment.

C. Design of APs and IoT devices

The prerequisite for supporting both RF and OWC is that IoT devices and APs must be equipped with the appropriate communication interfaces. RF interfaces consist of a dedicated transceiver and antenna; the VLC transmitter is typically an Light-Emitting Diode (LED) (often repurposed from illumination), and the receiver is a photodetector. In IR systems, a dedicated Infrared LED serves as the transmitter, paired with an IR-sensitive photodetector as the receiver.

Supporting hybrid RF/OWC connectivity mandates the integration of specific physical-layer components on IoT devices,

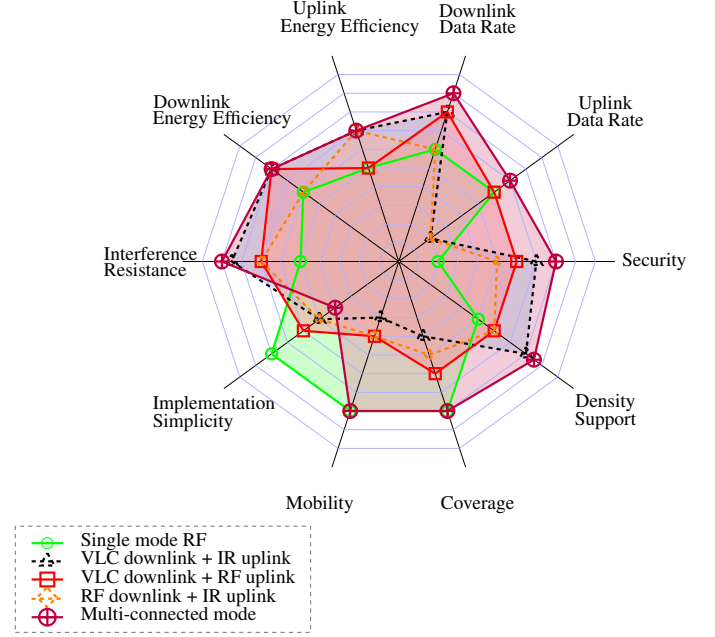


Figure 2. Performance comparison of different connectivity modes for hybrid OWC/RF IoT networks. For all metrics, a higher score indicates a more desirable outcome, with values increasing from the center (low/worst) to the perimeter (high/best).

including an Infrared LED (for the uplink) and a photodetector (for the downlink), alongside their existing RF interfaces. Although many devices contain simple photodiodes, these are typically for low-speed ambient-light sensing and lack the necessary high-bandwidth electronics for high-speed VLC demodulation. A more attractive alternative is the use of photovoltaic (PV) cells when device size permits, as they function as large-area photodiodes capable of simultaneously detecting data and harvesting energy. Since PV cells are often present in energy-harvesting designs, integrating the VLC downlink simply requires adding a low-power amplifier and processing circuitry, offering a low-cost path to incorporating OWC into resource-constrained IoT platforms.

In dedicated RF-only and OWC-only zones, the infrastructure is straightforward, relying on separate, specific APs designed for a single communication modality. However, the establishment of a hybrid zone—where both technologies are active—presents a fundamental architectural choice for the AP design. Hybrid zone can be covered in two distinct ways: either by utilizing separate APs for each modality (RF and OWC), or by deploying a single, integrated hybrid AP.

The first approach relies on two parallel infrastructures: using separate single-mode RF and single-mode OWC APs to cover the whole zone. While technically straightforward, this method increases hardware, cabling, and power costs. More critically, coordinating between physically separate APs for load balancing or handovers requires backhaul communication, which introduces latency and limits the system's ability to adapt in real time.

In contrast, integrated hybrid RF/OWC APs that house both modules in a single physical unit use a shared controller to manage both the RF and OWC interfaces through a unified

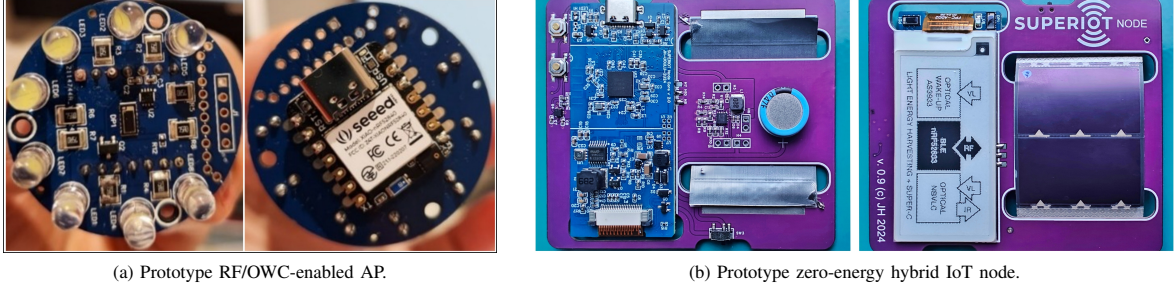


Figure 3. Hardware components developed within the EU SUPERIOT project [2], demonstrating the practical realization of the hybrid RF/OWC concept.

backhaul structure. This integration minimizes hardware redundancy and simplifies deployment while enabling superior real-time coordination. It facilitates intelligent load balancing, adaptive resource allocation, and energy-efficient operation that separate systems cannot easily match.

This framework is not merely conceptual; its practical feasibility has been demonstrated in the EU SUPERIOT project [2]. Figure 3a illustrates the project's prototype multi-modal RF/OWC-enabled AP, while Figure 3b shows the prototype of a zero-energy hybrid IoT node, powering all its communication functions entirely from harvested energy.

IV. HYBRID AP DEPLOYMENT STRATEGIES

To fully realize the benefits of hybrid RF/OWC systems, particularly the advanced functionalities of multi-connectivity and seamless failover, the IoT environment requires a properly dimensioned overlapping coverage area from both RF and OWC technologies, especially within the defined hybrid zone. The physical deployment of APs is the practical enabler for this hybrid coverage. However, AP deployment for a hybrid zone is challenging, as it must meticulously account for the distinct and often asymmetric propagation properties of each modality.

A. The Distinct Propagation Profiles

A central challenge in AP deployment for a hybrid zone stems from the intrinsic mismatch between RF and OWC propagation properties and their resulting coverage geometries. RF signals typically create a spherical or ellipsoidal coverage footprint, although the real-world coverage is shaped by antenna characteristics and environmental obstacles. In practice, this RF coverage—particularly in the sub-6 GHz frequencies common for IoT—is decently robust to such obstructions, allowing devices without a direct LoS to the AP to maintain connectivity.

In contrast, OWC relies mostly on directional LoS transmission, producing a confined conical beam that forms a circular footprint on the illumination plane. This radius depends directly on the AP mounting height h and the half-power angle θ , approximated by $h \times \tan(\theta)$. Any obstruction causes complete link loss, making OWC coverage far more sensitive to geometric placement and environmental conditions.

B. Strategies for Coverage Equivalence

The disparate coverage profiles of RF and OWC complicate the creation of a uniform hybrid zone. Using separate, single-mode OWC and RF APs simplifies the deployment problem, as it splits the challenge into two uncoupled optimization tasks and allows for independent placement. Conversely, integrated hybrid APs lack this inherent spatial flexibility because the single unit generates two disparate coverage patterns. Consequently, placement optimization for one interface is rarely optimal for the other, demanding meticulous planning to ensure balanced RF/OWC coverage.

A hybrid RF/OWC AP architecture comprises a central controller that manages both RF and OWC interfaces. Given that most indoor spaces already feature multiple LED deployments for illumination, the hybrid RF/VLC AP can leverage this existing infrastructure. This allows a single controller to manage multiple LEDs within a zone. By treating these LEDs as controllable optical transmitters, the controller can expand the optical footprint to better align with the RF coverage region.

The number of LEDs that must be coordinated to match the RF coverage area depends on several factors. In an ideal, blockage-free scenario, this quantity is primarily determined by the installation height and the RF interface's coverage radius. Figure 4 quantifies this relationship, showing that while higher LED mounting heights expand optical coverage and reduce the number of LEDs needed, the required count grows sharply with an increasing RF range. At greater ceiling heights or for shorter RF ranges, however, the LED count decreases substantially, making this coordinated hybrid coverage strategy highly feasible. Table I further demonstrates this relationship across common RF technologies. While Zigbee can be matched with relatively few LEDs, achieving coverage comparable to 2.4 GHz Wi-Fi demands dense LED deployments for VLC. The results also show that this requirement can be mitigated, as wider LED beam angles and greater installation heights both reduce the required LED count.

C. Hierarchical Control and Trade-offs

Coordinating multiple LEDs under a unified controller introduces a powerful architectural opportunity for hybrid RF/OWC networks. A shared control plane streamlines cabling and synchronization, enables centralized optimization of radio and optical resources, and allows lighting, connectivity, and energy management to be jointly orchestrated. As modern

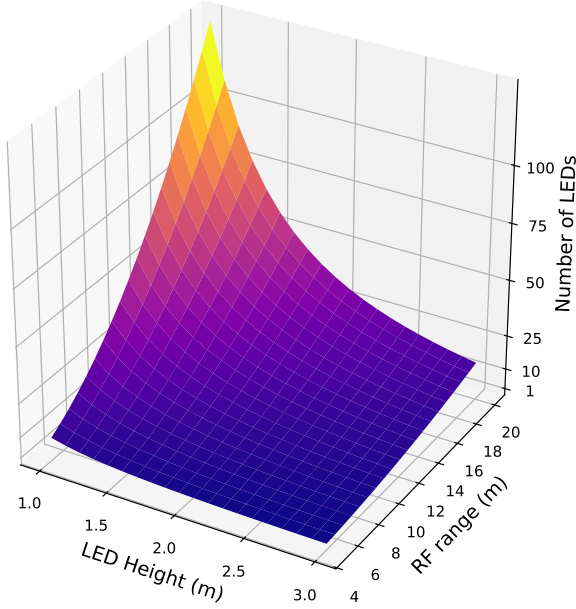


Figure 4. Required LED count to match RF AP coverage, by LED installation height and RF coverage range.

buildings increasingly adopt smart-lighting infrastructures — where LEDs are already networked for sensing, automation, and energy efficiency — the illumination system itself can evolve into a spatially distributed communications layer. In such environments, light becomes both a utility and a data channel, improving infrastructure reuse and overall system sustainability. Yet this integrated approach also introduces meaningful trade-offs, leading to the emergence of two major engineering challenges.

First, expanding the optical coverage footprint under a single controller simplifies deployment but introduces a fundamental trade-off: it dilutes the advantages of OWC. In practice, wider cells weaken physical-layer security, localization precision, and diminish scalability. Centralized control also creates a single point of failure. To resolve this trade-off, a hierarchical coordination architecture can be employed that operates analogously to a cellular network. In this model, a global controller acts as the core orchestrator, managing system-wide policies. The hybrid APs then function as distributed “edge controllers,” each managing its own local cluster of optical transmitters. This distributed structure allows each edge controller to maintain tightly confined light cells, thereby preserving the fine-grained OWC benefits (like device-level localization and zone-specific access control). Simultaneously, the global controller ensures coordinated system behavior across all APs and provides graceful degradation—the failure of a single edge AP does not bring down the entire network, solving the single-point-of-failure problem.

Second, hybrid deployment is not simply a matter of aligning RF and optical footprints; it requires the simultaneous consideration of multiple, often-conflicting system requirements. For instance, LEDs must primarily satisfy illumination requirements, which vary dynamically across different spaces

and times of day. Achieving consistent lighting quality and seamless hybrid coverage within a single physical infrastructure, therefore, calls for a co-design effort spanning the communication, lighting, and control domains. Furthermore, OWC performance is intrinsically dependent on the environment’s physical layout and application demands; factors such as furniture, partitions, user movement, and device orientation critically affect optical link quality, creating highly variable connectivity zones that behave fundamentally differently from RF systems. Ensuring robust service requires strategically overlapping optical cells, necessitating a higher degree of redundancy and completely different planning principles than those used for traditional RF cellular coverage. Thus, the simultaneous consideration of lighting quality, LoS geometry and blockage, distinct overlap requirements, mobility support, and energy budgets transforms deployment planning from a simple geometrical exercise into a complex, multi-objective optimization problem.

To overcome the challenges of integration and deployment, the widespread adoption of hybrid RF/OWC networks must be supported by advanced enabling technologies. These solutions require a unified, adaptive approach integrating Digital Twin (DT) modeling, Software-Defined Networking (SDN) control, and Artificial Intelligence (AI)-driven autonomy. The use of DTs creates high-fidelity virtual models of the environment to accurately simulate and predict complex interactions between illumination, link availability, and user mobility. This enables rigorous pre-deployment planning that minimizes coverage gaps and significantly enhances overall network observability. Concurrently, SDN control ensures essential programmability by dynamically allocating and steering traffic to the optimal channel in real-time based on instantaneous quality and priority. This capability accelerates recovery actions and acts as an active resilience lever for the hybrid network. Finally, AI-driven approaches are crucial for learning and adapting to the non-static hybrid environment, continuously optimizing network parameters—such as AP transmission power, coverage boundaries, and link aggregation strategies—in real-time to transition operations from reactive to proactive and anticipatory management.

V. CONCLUSION: HYBRID WIRELESS AS A PILLAR FOR RESILIENT 6G IOT

Hybrid RF/OWC networking represents a pivotal architectural evolution for meeting the 6G era’s simultaneous demands for massive capacity, intrinsic security, and high energy efficiency across dense and diverse IoT ecosystems. This article demonstrated how combining the ubiquity of RF with the spatial precision of OWC creates an adaptive wireless fabric that overcomes the inherent limitations of single-technology solutions for IoT ecosystems. We presented a holistic architecture for hybrid RF/OWC systems tailored to IoT environments and identified five distinct communication modalities. A comparative evaluation across ten performance metrics demonstrated the superior flexibility and robustness of multi-connectivity operation. We also argued that a unified hybrid AP architecture is key to minimizing cost and implementation complexity

Table I
ESTIMATED LED COUNT TO ACHIEVE VLC COVERAGE PARITY WITH DIFFERENT RF TECHNOLOGIES.

RF Technology	Typical Range (m)	h = 2.5 m		h = 3.0 m		h = 3.5 m	
		$\theta = 60^\circ$	$\theta = 80^\circ$	$\theta = 60^\circ$	$\theta = 80^\circ$	$\theta = 60^\circ$	$\theta = 80^\circ$
BLE	10 - 40	6 - 86	1 - 8	4 - 60	1 - 6	3 - 44	1 - 5
Zigbee	10 - 30	6 - 48	1 - 5	4 - 34	1 - 4	3 - 25	1 - 3
Wi-Fi (2.4 GHz)	45 - 50	108 - 134	11 - 13	75 - 93	7 - 9	56 - 69	6 - 7
Wi-Fi (5 GHz)	15 - 30	12 - 48	2 - 5	9 - 34	1 - 4	7 - 25	1 - 3

while enabling real-time cross-domain coordination in hybrid RF/OWC networks. A central practical challenge—coverage mismatch between RF and OWC—was examined through a quantitative coverage-equivalence analysis. The results provide actionable guidance on coordinated LED densities required to align optical footprints with typical RF coverage, reinforcing the technical and economic feasibility of integrated deployments. Looking forward, the evolution of hybrid RF/OWC networks will benefit from coordinated design frameworks that jointly consider communication, illumination, and control requirements. Emerging tools—such as DT, SDN-enabled control mechanisms, and AI-driven approaches—can support this direction by enabling more transparent planning, real-time adaptation, and efficient resource orchestration. These capabilities will play an important role in simplifying deployment, ensuring consistent coverage, and strengthening the reliability needed for broad commercial adoption.

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Mohammad Khalili is a doctoral researcher at the University of Oulu, focusing on heterogeneous optical wireless and RF networks, AI, and sustainable IoT.

Amila Perera is a doctoral researcher at the University of Oulu, focusing on Visible Light Communication, zero-energy, and sustainable IoT.

Juha Häkkinen is a Senior Research Fellow at the University of Oulu. His research focuses on circuits and systems for VLC and emerging circuit technologies utilizing novel flexible components in the IoT.

Marcos Katz is a Professor at the University of Oulu, specializing in wireless communications. He has held roles at Nokia, Samsung, and VTT, the Technical Research Centre in Finland. The major focus of his research is on the Visible Light Communication and sustainable IoT systems.

Konstantin Mikhaylov is an Associate Professor (Tenure Track) with the Centre for Wireless Communications at the University of Oulu. The major focus of his research is on the radio access and beyond-access technologies for massive and dependable IoT, and the matters related to the design and use of IoT systems.